



MEHLMANMEDICAL
HY USMLE REVIEW
PART I



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HY USMLE Review – Part I

Travel + self-limiting watery or brown/green diarrhea → Traveler diarrhea = ETEC HL or HS toxin

Bloody diarrhea + poultry consumption → *Campylobacter jejuni* or *Salmonella* spp.

Abx (clindamycin, beta-lactam, cephalosporin) + diarrhea → *C. difficile*

Dx of *C. diff* → stool AB toxin test, **not** stool culture

Fever of 104 + abdo distension in *C diff* → toxic megacolon → laparotomy

Tx of *C. diff* → **vancomycin, not metronidazole (updated guidelines as of Feb 2018)**

Bloody diarrhea + travel → *Entamoeba histolytica*

Tx of *E. histolytica* → metronidazole + iodoquinol; can give paromomycin

Close quarters or military barracks or cruise ship + watery diarrhea → Norwalk virus

Child <5 years + watery diarrhea → rotavirus

Few organisms causing bloody diarrhea → *Shigella*

Bloody diarrhea + appendicitis-like pain (pseudoappendicitis) in a child → *Yersinia enterocolitica*

Bloody diarrhea + reactive arthritis in an adult → *Y. enterocolitica*, *Campylobacter*, *Shigella*, *Salmonella*

Diarrhea + Guillain-Barre syndrome → *Campylobacter*

GBS syndrome CSF? → albuminocytologic dissociation (high protein + normal cells)

GBS syndrome Dx? → electromyography + nerve conduction studies (on NBME)

Cardiac ischemia + need to evaluate → ECG stress test first-line

Cardiac ischemia + abnormal baseline ECG (e.g., BBB) → Echo stress test (need normal ECG to do ECG stress test)

Cardiac ischemia + patient can't exercise → dobutamine + ECG/echo

ECG shows diffuse ST-segment elevations → pericarditis

Pericarditis Tx → NSAID, or steroid, or colchicine

Central chest pain worse when supine; better when leaning forward → pericarditis

Lateral chest pain after viral infection + increased CK → pleurodynia (intercostal muscle spasm)

ST-segment depressions in the anterior ECG leads → posterior MI

Electrical alternans on ECG → pericardial tamponade / pericardial effusion

Pulsus paradoxus (drop in systolic BP >10 mm with inspiration) → cardiac tamponade or severe asthma

Beck triad → hypotension + muffled heart sounds + JVD → pericardial tamponade

Tx of tamponade → pericardiocentesis or pericardial window

Tamponade → do echo before pericardiocentesis if both listed (even though sounds wrong, on 2CK NBME)

Community-acquired pneumonia (CPP) + bilateral CXR infiltrates → Mycoplasma

CPP + lobar pattern (right-lower lobe consolidation + dullness to percussion) → Strep pneumo

CPP + lobar pattern, but they say “interstitial” in the vignette description → Mycoplasma, not S. pneumo

Empiric Tx for CPP → azithromycin

Tx for CPP is pt on Abx past three months → fluoroquinolone over azithro

Pneumonia in CF patient <10 years → S. aureus exceeds Pseudomonas

Pneumonia in CF patient >10 years → Pseudomonas exceeds S. aureus.

Pneumonia after influenza infection → USMLE wants S. aureus

Pneumonia + rabbits → F. tularensis

Pneumonia + cattle → Coxiella (Q fever)

Pneumonia + birds → Chlamydia psittaci

Leg swelling + pain + shortness of breath → Pulmonary embolism caused by DVT

Tx of PE → Heparin before spiral CT

Tx of PE in pregnant woman → V/Q scan, not CT

Tx of PE in someone already on warfarin → spiral CT to confirm, then IVC filter

Acid-base disturbance in PE → resp. alkalosis (low CO₂, high pH, low O₂, normal bicarb [too acute to change])

Acid-base disturbance in asthma → resp. alkalosis (low CO₂, high pH, low O₂, normal bicarb [too acute to change])

Acid-base disturbance in aspirin toxicity first 20 mins → resp. alkalosis (low CO₂, high pH, normal O₂, normal bicarb [too acute to change])

Acid-base disturbance in aspirin toxicity after 20 mins → mixed metabolic acidosis-respiratory alkalosis (low CO₂, low pH, normal O₂, low bicarb)

Tx for aspirin toxicity → bicarb (increased excretion through urinary alkalinization)

Tx for TCA toxicity → sodium bicarb → dissociates drug from myocardial sodium channels

Normal calcium → 8.4-10.2 mEq/L

Tx of hypercalcemia → 10.2-12 → normal saline (0.9% NaCl)

Tx of hypercalcemia → 12-14 → normal saline (0.9% NaCl) only if asymptomatic; add bisphosphonate (e.g., pamidronate) if symptomatic

Tx of hypercalcemia → 14+ → normal saline (0.9% NaCl) + bisphosphonate

High calcium + polyuria → nephrogenic diabetes insipidus (weird, but on NBME)

High calcium + confusion → delirium caused by high calcium

Low calcium or potassium not responsive to supplementation → cause is low Mg

Low calcium or potassium in alcoholic → hypomagnesemia is cause

Ataxia, confusion, ophthalmoplegia → Wernicke encephalopathy (B1 deficiency)

Retrograde amnesia + confabulation in alcoholic → Korsakoff psychosis

Wernicke-Korsakoff syndrome → mammillary bodies

Hx of many pregnancies + downward movement of vesicourethral junction → stress incontinence

Tx of stress incontinence → pelvic floor exercises (Kegel); if insufficient → mid-urethral sling

Hyperactive detrusor or detrusor instability → urge incontinence

Need to run to bathroom when sticking key in a door → urge incontinence

Incontinence in multiple sclerosis patient or perimenopausal → urge incontinence

Tx of urge incontinence → oxybutynin (muscarinic cholinergic antagonist) or mirabegron (beta-3 agonist)

Incontinence + high post-void volume (usually 3-400 in question; normal is <50 mL) → overflow incontinence

Incontinence in diabetes → overflow incontinence due to neurogenic bladder

Tx for overflow incontinence in diabetes → bethanacol (muscarinic cholinergic agonist)

Incontinence in BPH → overflow incontinence due to outlet obstruction → eventual neurogenic bladder

Tx for overflow incontinence in BPH → insert catheter first; then give alpha-1 blocker or 5-alpha-reductase inhibitor; then TURP if necessary

Exquisitely tender prostate on digital rectal exam → prostatitis

Prostatitis Tx → ciprofloxacin (fluoroquinolone → DNA gyrase / topoisomerase II inhibitor)

Costovertebral angle tenderness + fever → pyelonephritis

Costovertebral angle tenderness + granular casts → pyelonephritis (correct, super-weird; NOT acute tubular necrosis; this is on 2CK NBME)

Tx for pyelonephritis → ciprofloxacin, OR ampicillin + gentamicin

Saddle anesthesia + urinary retention → cauda equina syndrome

Perianal anesthesia + urinary retention or incontinence → conus medullaris syndrome

Gradual-onset dementia + no sensory or motor dysfunction → Alzheimer

Mini-mental state exam score low + patient tries to do well → Alzheimer

Mini-mental state exam score low + patient is apathetic / takes long to perform tasks / does poorly on reverse serial 7s → depression (pseudodementia)

Patient complains about memory loss → normal aging, not Alzheimer

First-line Tx for Alzheimer → donepezil (cholinesterase inhibitor → increases cholinergic transmission); can also give galantamine or rivastigmine

NMDA receptor (glutamate receptor) antagonist used in Alzheimer Tx → memantine

Step-wise dementia/decline and/or sensory/motor disturbance → vascular dementia

Hx of hypertension + dementia + sensory/motor disturbance → vascular dementia

Visual hallucinations + Parkinsonism + dementia → Lewy body dementia

Apathy + disinhibition + personality change + dementia → frontotemporal dementia (Pick disease)

Urinary incontinence + ataxia + CNS dysfunction (wet, wobbly, wacky) → Normal-pressure hydrocephalus

Wet, wobbly, wacky + Parkinsonism → still NPH

Parkinsonism in young patient → Wilson disease till proven otherwise

Parkinsonism in older patient → Parkinson disease

Parkinsonism + axial dystonia → progressive supranuclear palsy

Tx of UTI → TMP/SMX or nitrofurantoin

Tx of cystitis → nitrofurantoin (need not be pregnant)

Waiter tip position in kid → upper brachial plexus injury → C5/6 → Erb-Duchenne palsy

Claw hand → lower brachial plexus → C8/T1 → Klumpke palsy

Pronated arm + wrist-drop → radial nerve injury

Midshaft fracture of humerus → Radial nerve injury

Supracondylar fracture of humerus → median nerve injury

Surgical neck of humerus fracture → axillary nerve injury

Medial epicondylar injury → ulnar nerve injury

Weakened biceps + loss of sensation of lateral forearm → musculocutaneous nerve injury

Paresthesias + pain following burn or casting → compartment syndrome

Compartment syndrome Dx → measure compartment pressure

Compartment syndrome Tx → fasciotomy of uncast

Guy lifts heavy box → severe lower back pain + muscle spasm + no radiculopathy → lumbosacral strain only

Lumbosacral strain diagnosis → DO NOT x-ray

Lumbosacral strain Tx → NSAIDs + exercise as tolerated; bedrest is the WRONG answer

Guy lifts heavy box → severe lower back pain + radiculopathy → herniated disc; yes, x-ray.

Point tenderness over a vertebra in older woman → osteoporosis (compression fracture)

Point tenderness over a vertebra in younger patient on steroids → osteoporosis (compression fracture)

Point tenderness over a vertebra in patient with autoimmune disease → recognize patient is on steroids → osteoporosis (compression fracture)

Point tenderness over a vertebra in IV drug user → epidural abscess

“Step-off” of one vertebra relative to another → spondylolisthesis

Back pain worse in the morning and gets better throughout day in male 20s-40s → ankylosing spondylitis

Bamboo spine → ankylosing spondylitis

Dx of AS → x-ray of sacroiliac joints

Back pain worse when standing or walking for long periods of time → lumbar spinal stenosis

Radiculopathy down an arm → cervical disc herniation

Bilateral paresthesias in the arms in rheumatoid arthritis patient → atlantoaxial subluxation

Bilateral paresthesias in the arms in rheumatoid arthritis patient → MR of spine to Dx atlantoaxial subluxation

Prior to surgery in rheumatoid arthritis patient → cervical CT or flexion/extension x-rays of cervical spine to check for atlantoaxial subluxation

Back pain in elderly patient with hypercalcemia → multiple myeloma or metastases

Back in pain in patient with history of other type of cancer → metastases

Suspected spinal mets → MRI

Metastases to long bones in prostate cancer → osteoblastic (Dx with bone scan); spine do MRI

High hemoglobin +/- pruritis after shower +/- plethora +/- splenomegaly → polycythemia vera

High hemoglobin + low EPO → polycythemia vera

Pruritis after shower → basophilia

High hemoglobin + lung disease / low pO₂ → secondary polycythemia (high EPO)

Polycythemia + hypercalcemia + smoker + red urine → RCC (paraneoplastic EPO + PTH-rp)

Blurry vision or Raynaud or pain in fingers or headache → hyperviscosity syndrome

Hyperviscosity syndrome → Waldenstrom macroglobulinemia or polycythemia

Hereditary spherocytosis → AD, ankyrin or spectrin or band protein deficiency; Tx = splenectomy

Treatment for ITP → steroids, then IVIG, then splenectomy

Tx for hereditary hemochromatosis → serial phlebotomy

Tx for secondary hemochromatosis (transfusional siderosis) → chelation therapy (deferoxamine)

Viral infection + tinnitus + vertigo +/- neurosensory hearing loss → labyrinthitis

Viral infection + vertigo → vestibular neuritis

Tx for multiple sclerosis flares → IV steroids (IV methylprednisolone)

Given to MS patients between flares → interferon-beta

Tx for spasticity in MS → baclofen (GABA-B receptor agonist)

Incontinence in MS → urge (hyperactive detrusor, as mentioned earlier)

New-onset murmur + fever → endocarditis till proven otherwise

Empiric Tx for endocarditis → vancomycin or ampicillin/sulbactam, PLUS gentamicin

Beta-lactam (e.g., nafcillin) or cephalosporin + rash + renal issue → interstitial nephritis

Interstitial nephritis → WBCs in the urine (eosinophils)

Fixed splitting of S2 → atrial septal defect

Holosystolic murmur at left sternal border PLUS either parasternal heave or palpable thrill → VSD

Holosystolic murmur at left sternal border PLUS diastolic rumble → also VSD

To-and-fro murmur → PDA (on 2CK NBME)

Pan-systolic-pan-diastolic murmur → PDA

Continuous, machinery-like murmur → PDA

Congenital rubella syndrome → PDA

Heart problem in neonate of mom with SLE → congenital third-degree heartblock

Heart problem in William syndrome → supravalvular aortic stenosis

Bicuspid aortic valve → aortic stenosis

Mid-systolic (crescendo-decrescendo) murmur + gets worse with Valsalva → HOCM

Mid-systolic (crescendo-decrescendo) murmur + no change or softens with Valsalva → aortic stenosis

Myxomatous degeneration of mitral valve → mitral valve prolapse

Marfan or Ehlers-Danlos syndrome → MVP or aortic regurg

Rheumatic heart disease acutely (onset of Group A Strep infection) → mitral regurg

Rheumatic heart disease later on (years after infection) → mitral stenosis

Mid-systolic click → MVP

Late-peaking systolic murmur with ejection click → another way they describe aortic stenosis

Bounding pulses + massively wide pulse pressure → aortic regurg

Brisk upstroke + precipitous downstroke of pulse → aortic regurg

Syncope + angina + dyspnea (SAD) → aortic stenosis

Dyspnea in second trimester of pregnancy → mitral stenosis

Dyspnea late in pregnancy → peripartum cardiomyopathy

Screening at age 50 → mammogram (every two years) + colonoscopy (every ten years)

Colon cancer in first-degree relative (sibling or parent) → start at age 40 or ten years before diagnosis in relative, whichever is earlier, and do every 5 years.

Breast imaging (if performed) → ultrasound only under age 30; over age 30 do mammogram +/- ultrasound

Anuria (no urine output) after removal of catheter → acute urethral obstruction

Contrast induced nephropathy; how to prevent → saline hydration beforehand

BUN/Cr > 20 → prerenal (hypovolemia) → heart failure

BUN/Cr < 20 → not prerenal (15-20 for post-renal, and <15 for intra-renal is wrong on 2CK NBMEs)

Blood loss + oliguria → acute tubular necrosis

Blood loss + obstetric catastrophe → diffuse cortical necrosis

Sickle cell + nephrotic syndrome → focal segmental glomerulosclerosis

Sickle cell + red urine → renal papillary necrosis

2+ blood in urine but 0-4 RBCs/HPF on LM → false + blood on dipstick → myoglobinuria → rhabdomyolysis

Pleural / supradiaphragmatic plaques → asbestosis

Preferred antibiotic in sepsis → ceftriaxone

Sepsis Tx in young children → cefotaxime

Tx for spontaneous bacterial peritonitis → ceftriaxone

Cirrhosis or recent peritoneal dialysis or nephrotic syndrome + fever + abdo pain → SBP

SBP Dx → paracentesis (peritoneal aspiration; don't confuse with pericardiocentesis) + do gram-stain + look for >250 WBCs per HPF

Dysphagia to solids and liquids at the same time to start → says neurogenic cause → achalasia

Dysphagia to solids that **progresses** to solids and liquids → esophageal cancer

Halitosis +/- gurgling sound when swallowing +/- regurgitation of undigested food → Zenker

Zenker + achalasia initial Dx modality → barium swallow

After barium swallow is done and shows bird's beak appearance → manometry to confirm Dx of achalasia

Pt with Hx of GERD + dysphagia → straight to endoscopy to rule out cancer

Diabetic pt with new-onset GERD → diabetic gastroparesis

Diabetic pt with new-onset GERD → give metoclopramide, not PPI

Diabetic gastroparesis before giving med → endoscopy first to rule out physical obstruction

Endoscopy negative for diabetic gastroparesis → do gastric-emptying scintigraphy

Bulimia nervosa or anorexia → never give bupropion (seizure risk)

Electrolyte abnormality in anorexia → hypokalemia

Most common cause of death in anorexia → arrhythmia from hypokalemia

Refeeding syndrome → worry about hypophosphatemia

Tx of anorexia + depression → mirtazapine (alpha-2 antagonist); stimulates appetite

Amenorrhea in anorexia → low FSH + low estrogen (hypogonadotropic)

Premature ovarian failure + Turner syndrome + menopause → high FSH (low inhibin) + low estrogen

Cholelithiasis → Dx with abdo ultrasound; fat, forties, female, fertile

Biliary colic + fever → cholecystitis

Tx of cholelithiasis + cholecystitis → cholecystectomy

Pt doesn't want surgery or is pregnant → ursodeoxycholic acid (ursodiol)

Abdo USS negative in suspected cholecystitis → HIDA scan

Gall bladder doesn't light up on HIDA scan → confirms cholecystitis

Cholelithiasis in pregnancy → estrogen upregulates HMG-CoA reductase + progesterone slows biliary peristalsis

Trichotillomania (eating one's hair) + GI symptoms → gastric bezoar (hair ball)

Hx of surgery + high-pitched bowel sounds or acute-onset abdo symptoms → small bowel obstruction (SBO)

Renal failure + friction rub over chest → uremic pericarditis → do hemodialysis

High leukocytes + high leukocyte ALP → leukemoid reaction

High leukocytes + low leukocyte ALP → CML

Metamyelocytes + myelocytes + splenomegaly → CML

Tx of CML → imatinib; causes fluid retention / edema

Smudge cells + autoimmune hemolytic anemia → CLL

Auer rods → AML; composed of myeloperoxidase

Tx of AML → DIC caused by Auer rod release into blood

Kid with high lymphocytes → ALL or pertussis (weird bc bacterial, but lymphocytes often >30k)

Dry cough in winter → cough-variant asthma (1/3 of asthmatics only have cough)

Young African American woman + dry cough + normal CXR → asthma (activation of mast cells), not sarcoidosis

Young African American woman + dry cough + nodularity on CXR → sarcoidosis (noncaseating granulomas)

Electrolyte disturbance in sarcoid → hypercalcemia → LOW PTH + high calcium

Hypercalcemia in sarcoid, why? → epithelioid (activated) macrophages produce 1-alpha hydroxylase, thereby activating vitamin D3

Increased calcium in sarcoid → means decreased calcium in feces (bc D3 increased small bowel absorption)

Tx for sarcoid → steroids

Outpatient Tx of asthma → SABA, then low-dose ICS, then maximize dose of ICS, then LABA, then use any number of drugs (e.g., mast cell stabilizers, anti-leukotriene, etc.), then oral steroids last resort

Kid with asthma on SABA inhaler + not effective + next best step? → ICS (fluticasone)

Kid with asthma on SABA inhaler + most effective way to decrease recurrence? → oral steroids (not next best step, but certainly most effective)

40s male + hematuria + hemoptysis → Goodpasture syndrome

Antibodies in Goodpasture → Anti-GBM (anti-collagen IV)

Dx of Goodpasture → antibodies first, but confirmatory is renal biopsy showing linear immunofluorescence

Hematuria + hemoptysis + "head-itis" (mastoiditis, sinusitis, otitis, nasal septal perforation) → Wegener

New name for Wegener → granulomatosis with polyangiitis

Dx of Wegener → c-ANCA (anti-PR3; anti-proteinase 3)

Asthma + eosinophilia → Churg-Strauss

New name for CS → eosinophilic granulomatosis with polyangiitis

Dx of CS → p-ANCA (anti-MPO; anti-myeloperoxidase)

Hematuria in isolation + p-ANCA in serum → microscopic polyangiitis (MP)

Severe renal disease in Wegener or Goodpasture or MP → rapidly progressive glomerulonephritis (crescentic)

High ALP + high direct bilirubin + high amylase or lipase → gallstone pancreatitis = choledocholithiasis

High ALP + high direct bilirubin + high amylase or lipase + remote Hx of cholecystectomy → sphincter of Oddi dysfunction (can't be a stone cuz the gallbladder was removed ages ago)

High ALP + high direct bilirubin + normal amylase or lipase in someone with recent cholecystectomy → choledocholithiasis (retained stone in cystic duct that descended, but not distal to pancreatic duct entry point)

Dx and Tx of choledocholithiasis → ERCP

High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy → pancreatic cancer

Dx of pancreatic cancer → CT abdo with contrast

High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy + CT is negative → cholangiocarcinoma

High ALP + high direct bilirubin + normal amylase or lipase + diffuse pruritis + high cholesterol → primary biliary cirrhosis (PBC)

High ALP + high direct bilirubin + normal amylase or lipase + autoimmune disease (in pt or family) → PBC

Dx of PBC → anti-mitochondrial Abx next best step; liver biopsy is confirmatory

Recent cholecystectomy + fever + abdo pain → post-op bile leak

High ALP + high direct bilirubin + normal amylase or lipase + CT shows cystic lesion in bile duct → choledochal cyst → do simple excision of cyst (cholangiocarcinoma not cystic + CT can be negative)

Imaging to view liver or pancreas → CT with contrast

Imaging to view gallbladder → Ultrasound

Imaging to view gallbladder in suspected cholecystitis only if USS negative → HIDA scan

Imaging to view bile ducts → ERCP or MRCP (choose ERCP > MRCP if both listed)

Teenage girl with Hx of cutaneous candida infections since childhood → chronic mucocutaneous candidiasis

MCC → answer = T cell dysfunction = impaired cell-mediated immunity on the USMLE

Bacterial + fungal + protozoal + viral infections since birth → SCID

Bacterial infections since age 6 months → Bruton

Bacterial infections only since birth → Bruton (rare as hell to say from birth, but it's on new 2CK NBME)

SCID XR variant → common gamma-chain mutation (IL-2 receptor deficiency)

SCID AR variant → adenosine deaminase deficiency

Bruton mechanism → tyrosine kinase mutation

Hyper IgM syndrome → deficiency of CD40 ligand on T cell (can't activate CD40 on B cell to induce isotype class switching)

Greasy, scaly scalp + itchy + papules + adult → seborrheic dermatitis

Tx for SD → azole or selenium shampoo

Tx for tinea capitis → oral griseofulvin for patient only

How to decrease risk of tinea capitis → avoidance of sharing of hats

Tx of onychomycosis (nailbed fungus) → oral terbinafine

Tx of tinea pedis → topical terbinafine or topical azole

Tx of tinea corporis (ring worm) → topical azole (clotrimazole or miconazole)

Tx of cutaneous candida → oral azole

Tx of oropharyngeal candida → nystatin mouthwash

Tx of esophageal candidiasis → oral azole, not nystatin mouthwash

Tx of vaginal candidiasis → topical nystatin before trying oral azole

Odynophagia (painful swallowing) in immunocompromised pt → esophageal candidiasis till proven otherwise

CNS fungal infection or fungemia (rigors/chills) → amphotericin B

Cryptococcal meningitis → amphotericin B + flucytosine, then do fluconazole taper

Simple fungal pneumonia → fluconazole

Sporothrix schenckii (rose thorn + finger papule) → itraconazole

Hypopigmentation on upper back / trunk → tinea versicolor (Malassezia furfur)

Tx of tinea versicolor → topical selenium

Most common cause of impetigo → S. aureus now exceeds S. pyogenes

Tx of impetigo → topical mupirocin

Beefy red, well-demarcated skin plaque → erysipelas

Most common cause of erysipelas → Group A Strep (S. pyogenes) >>> S. aureus

More diffuse pink skin lesion + tenderness + fever → cellulitis

Most common cause of cellulitis → *S. aureus* exceeds *S. pyogenes*

Tx of erysipelas + cellulitis → oral dicloxacillin or oral cephalexin

Wide-complex tachycardia → ventricular tachycardia (VT)

Narrow-complex tachy → SVT

Tx for SVT → vagal/carotid massage first; if doesn't work, then adenosine

Tx of VT → anti-arrhythmics, e.g., amiodarone

Tx of SVT or VT in setting of coma / unconsciousness → direct-current countershock

Tx of first-degree heartblock or second-degree Mobitz I (Wenckebach) → observe

Tx of second-degree Mobitz II or third-degree heartblock → pacemaker

First-degree heartblock → PR-interval >200 milliseconds

Mobitz I → gradually prolonging PR-interval before a QRS drops

Mobitz II → no gradual prolongation of PR-interval; QRS randomly drops

Third-degree → HR super slow at 30-40; no relation between p-waves and QRS complexes

Infective causes of third-degree → Lyme disease, congenital lupus, diphtheria

Give killed IM influenza vaccine when? → Every year in fall/winter only; start from 6 months of age

Killed IM Influenza vaccine safe in pregnancy? → Yes, give anytime to pregnant women

Live-attenuated intranasal influenza vaccine guidelines? → Only give age 2-49 to non-pregnant, non-immunocompromised persons

Vaccines at age 2, 4, 6 months: HepB, Polio Salk, Pneumo PCV13, DPT, HiB, rotavirus (also give HepB at birth)

HPV vaccine when → age 9-45

Mom's HepB status unknown → give neonate HepB vaccine; only give immunoglobulin if mom comes back +

MMR → first dose at 12-15 months; second dose age 4-6 years

Varicella → one dose between 12-18 months

Age 65 or older → give Pneumo PCV13 followed by PPSV23 6-12 months later

Asplenia or sickle cell → PCV13 + PPSV23 + HiB + Meningococcal

Circular lesion in pancreas seen in pancreatitis → pseudoabscess → answer = ERCP to drain internally

Bullous changes on CXR or expanded lungs / hyperlucency → emphysema

Centri-acinar emphysema → smokers

Pan-acinar emphysema → alpha-1 anti-trypsin deficiency

Young adult + non-smoker + has emphysema + relative died of hepatic cirrhosis → alpha-1 anti-trypsin deficiency

CREST syndrome lung pathology? → can cause pulmonary fibrosis → pulmonary hypertension

Restrictive lung disease → normal or increased FEV1/FVC

Obstructive lung disease → decreased FEV1/FVC

Why is FEV1/FVC normal or high in restrictive? → radial traction on outside of airways is sticky (keeps airways from closing)

Apex to base lung changes when sitting/standing → **both** ventilation + perfusion increase apex to base

Most common cause of otitis media → Strep pneumo

Tx of otitis media → oral amoxicillin only

Tx of recurrent OM → amoxicillin/clavulanate

When to do tympanostomy tube → three or more OM in 6 months, or 4 or more in a year

Most common cause of otitis externa → Pseudomonas

Tx of otitis externa → topical ciprofloxacin + hydrocortisone drops

Prevention of OE in someone with constant water exposure (e.g., crew team) → alcohol-acetic acid drops

Tx of earwax buildup → carbamide peroxide drops

Low hematocrit + low MCV + low transferrin + low TIBC + transferrin saturation normal or low → anemia of chronic disease

Low hematocrit + low MCV + high transferrin + high TIBC + transferrin saturation super-low → iron deficiency anemia

Low hematocrit + low MCV + increased red cell distribution width (RDW) → iron deficiency anemia

Low hematocrit + low MCV + low/low-normal RDW → thalassemia

Low hematocrit + low MCV + low iron + low ferritin → iron deficiency

Low hematocrit + low MCV + normal iron + normal or high ferritin → thalassemia

Low hematocrit + low MCV + normal iron + normal ferritin in pregnant woman on iron supplements → thalassemia

Microcytic anemia that doesn't improve with iron supplementation → thalassemia

Dx of thalassemia → hemoglobin electrophoresis

Low hematocrit + normal MCV + low iron + normal or high ferritin → anemia of chronic disease

Tx of anemia of chronic disease if renal failure is cause → answer = EPO

Tx of anemia of chronic disease if renal failure not cause (IBD, RA, SLE, etc.) → CANNOT give EPO; Tx underlying condition.

High BP + smoker + TIA or stroke or retinal artery occlusion. How to best decrease stroke risk → Answer = lisinopril, not smoking cessation

Normotensive old pt + TIA or stroke or retinal artery occlusion → atrial fibrillation

Hypertensive pt + stroke → do carotid duplex ultrasound

Normotensive pt + stroke → do ECG; if ECG normal → Holter monitor

High BMI female + irregular menstrual cycles → anovulation

Anovulation + hirsutism → PCOS

Anovulation. Cause USMLE wants? → insulin resistance → causes abnormal GnRH pulsation

Why hirsutism in anovulation → abnormal GnRH pulsation causes high LH/FSH ratio

Why high LH/FSH ratio important in anovulation/PCOS → ovulation stimulated when follicle not ready → no ovulation (anovulation) → follicle retained as cyst

What's LH do? → Stimulates theca interna cells (females) and Leydig cells (males) to make androgens

What's FSH do? → Stimulates granulosa cells (females) and Sertoli cells (males) to make aromatase; also primes follicles

Tx for PCOS → if high BMI, weight loss first always on USMLE

Tx for PCOS if they ask for meds and/or weight loss already tried → OCPs (if not wanting pregnancy); clomiphene (if wanting pregnancy)

PCOS increases risk of what → endometrial cancer (unopposed estrogen)

Tx of prostate cancer → flutamide + leuprolide together (if they force a sequence, choose F then L).

Tx of acute gout → indomethacin (NSAID) first on USMLE; then steroids, then colchicine

Tx of acute gout if indomethacin + steroids not listed → colchicine

Tx of acute gout in pt with renal insufficiency or Hx of renal transplant → steroids

Tx of chronic gout (decrease recurrence) → allopurinol or febuxostat (xanthine oxidase inhibitors)

Never give which drug to pt with Hx of uric acid stones or over-producer → probenecid (uricosuric)

What are rasburicase / pegloticase → urate oxidase analogues → cleave uric acid directly

Young kid + self-mutilation + red-orange crystals in diaper → Lesch-Nyhan syndrome (X-linked)

Lesch-Nyhan enzyme → HGPRT deficiency

Crystal type in pseudogout → calcium pyrophosphate deposition disease

Two main causes of pseudogout → primary hyperparathyroidism + **hemochromatosis**

Two ways pseudogout presents → monoarthritis of large joint (i.e., knee) or **osteoarthritis-like presentation in someone with primary hyperparathyroidism or hemochromatosis**

32M + dark skin on forearms + increased fasting glucose; Dx? → hemochromatosis (bronze diabetes)

Same male + painful hands + x-ray shows DIP involvement. Joint pain Dx? → **pseudogout**

Tx of pseudogout → same as gout acutely; Tx underlying condition for chronic

Biggest risk factor for osteoarthritis → obesity

Tx of osteoarthritis → weight loss; if normal BMI → acetaminophen before NSAIDs

Patient with OA taking naproxen (NSAID) + peripheral edema → increased renal retention of sodium

Patient taking NSAID + edema; why? → NSAID decreases renal blood flow → PCT increases Na reabsorption to compensate for perceived low volume status → water follow sodium

Tx of rheumatoid arthritis → Two-armed: symptom-relief + disease-modifying anti-rheumatic drugs (DMARDs)

Symptom-relief for RA → NSAID first, then steroids (these do symptoms only; do not slow disease progression)

DMARDs for early RA → always methotrexate first; if insufficient, add another DMARD (sulfasalazine or leflunomide); if insufficient add anti-TNF-alpha agent

Methotrexate MOA → dihydrofolate reductase inhibitor

Methotrexate side-effects → pulmonary fibrosis + hepatotoxicity + mouth ulcers (neutropenia)

Sulfasalazine MOA → metabolized into sulfapyridine + mesalamine in the gut by bacteria

Mesalamine is 5-ASA absorbed as the Tx for RA; only NSAID considered to be DMARD

Leflunomide MOA → dihydroorotate dehydrogenase inhibitor (pyrimidine synthesis)

Most specific Abs in RA → anti-CCP (cyclic citrullinated peptide), not RF (rheumatoid factor)

X-ray of hands in RA vs OA → Only OA has DIPs involved; RA is PIPs + MCPs

Symmetry in RA vs OA → RA is symmetrical; OA is not

Any pt with red, warm, tender knee → joint aspiration (arthrocentesis); septic arthritis till proven otherwise

Biggest risk factor for septic arthritis → abnormal joint architecture

Pt groups most likely to get SA → prosthetic joints, RA/OA, recent intense exercise/joint trauma; peds (JRA)

Pt group most likely to get SA → those with prosthetic joints (can't be more abnormal than fake joint)

Pt with OA or RA has red, warm, tender knee → do arthrocentesis (septic arthritis)

17F had kickboxing tournament last weekend + knee is red, warm, tender → arthrocentesis (SA)

Kid + recurrent knee redness, warmth, pain + fever → Juvenile rheumatoid arthritis (JRA; Still disease)

Kid + recurrent joint pain +/- high ESR +/- rash → JRA

Kid + recurrent joint pain + anemia → JRA (anemia of chronic disease)

Kid with suspected JRA has sore knee → must do arthrocentesis to rule out septic arthritis

Most common presentation finding in SLE → arthritis (>90%)

Woman 20s-40s + arthritis + thrombocytopenia → SLE

Woman 20s-40s + arthritis + mouth ulcer + circular skin lesions → SLE

Malar rash + low RBCs + low WBCs + low platelets; mechanism for low cell lines? → increased peripheral destruction (antibodies against hematologic cells lines seen in SLE; isolated thrombocytopenia most common)

Tx of SLE flare → steroids

SLE + red urine; Dx? → lupus nephritis, more specifically, diffuse proliferative glomerulonephritis (DPGN)

Histology of DPGN → wire looping capillary pattern

Tx of lupus nephritis → mycophenolate mofetil

Tx of discoid lupus → hydroxychloroquine

Most specific Abs for SLE → anti-Smith (RNP), not anti-dsDNA

Which Abs go up in acute SLE flares → anti-dsDNA (and C3 goes down)

Drug-induced lupus Abs → anti-histone

Drugs that cause DIL → Mom is HIPP → Minocycline, Hydralazine, INH, Procainamide, Penicillamine

Viral infection + all three cell-lines are down → viral-induced aplastic anemia

Viral-induced aplastic anemia; next best step in Dx? → bone marrow aspiration

Viral-induced aplastic anemia; mechanism? → defective bone marrow production (contrast with SLE)

Viral infection + low platelets → ITP (immune thrombocytopenic purpura)

Woman 30s-40s with random bruising at different stages of healing → (also ITP; first rule out abuse)

Mechanism of ITP → Abs against GpIIb/IIIa on platelets

Dx of ITP → answer = low platelet count; don't choose increased bleeding time

ITP Tx → steroids first, then IVIG, then splenectomy

ITP episode → next best step in management → steroids

ITP episode → most effective way to decrease recurrence → splenectomy (not first-line, but most effective)

Family Hx of heme condition treated with splenectomy → hereditary spherocytosis (autosomal dominant)

Bleeding time meaning? → platelet problem

PT and aPTT meaning? → clotting factor problem

Heme findings in ITP → increased BT, normal PT, normal aPTT

Heme findings in hemophilia → increased aPTT; bleeding time and PT are normal

Cause of hemophilia → X-linked recessive; hemophilia A (factor VIII def); hemophilia B (factor IX def)

Tx of hemophilia A → desmopressin for hemophilia A (increases VIII release); then give factor VIII

Tx of hemophilia B → give factor IX

Classic hemophilia presentation → hemarthrosis in school-age boy; bleeding after circumcision in neonate

Inheritance pattern of vWD → AD

Heme findings in vWD → bleeding time always high; PT always normal; aPTT elevated half the time

What is main function of vWF? → bridges platelet GpIb to underlying collagen (adhesion, not aggregation)

What is secondary function of vWF → stabilizes factor VIII in plasma (that's why aPTT only half time increased)

vWD presentation → always one platelet problem + one clotting factor problem

Platelet problem? → epistaxis, bruising, petechiae → generally mild and cutaneous

Clotting factor problem → menorrhagia, excessive bleeding with tooth extraction, hemarthrosis (but

hemarthrosis very very rare in vWD; it is seen in hemophilia)

vWD treatment → desmopressin → increases release of vWF

Vitamin K deficiency heme parameters? → Increased PT + aPTT; bleeding time normal

Cause of vitamin K deficiency in adults → chronic Abx knock out colonic flora

Cause of sickle cell → glutamic acid to valine mutation on beta-chain

Inheritance of sickle cell → AR

Nephrotic syndrome in SS → FSGS

Dark urine in SS → renal papillary necrosis

HY drugs that cause agranulocytosis → clozapine, ganciclovir, propylthiouracil, methimazole, methotrexate, ticlopidine

How will agranulocytosis (neutropenia) present on USMLE? → mouth ulcers + fever

Tx for febrile neutropenia / neutropenic fever → immediate broad-spectrum IV Abx

Broad-spectrum Abx example? → Piperacillin/tazobactam; cefepime + vancomycin

What is ganciclovir used for? → Tx of CMV (DNA polymerase inhibitor)

PTU and methimazole are used for what? → Tx of Graves

Ticlopidine is what? → ADP2Y12 blocker anti-platelet agent (clopidogrel, prasugrel, ticagrelor also)

Strongest indication for anti-coagulation → prosthetic material in heart / prosthetic valve (factoid in isolation)



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MEHLMANMEDICAL HY USMLE REVIEW PART I

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